Unit no. ° 12

COORDINATE GEOMETRY: REVIEW OF VECTORS AND LINES

15-26 March 2027

I Parallelogram and intersection

Example 12.1: Booklet review

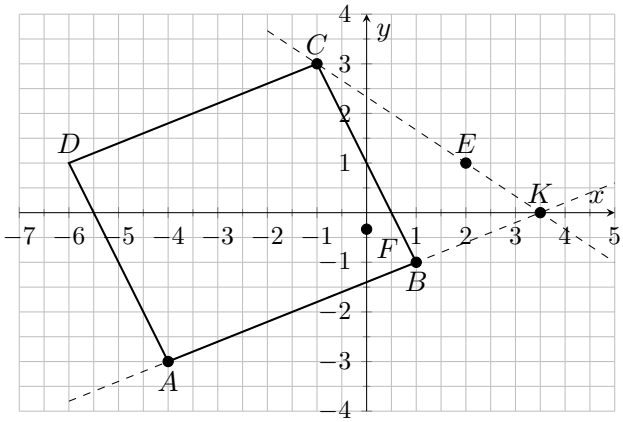
In an orthonormal coordinate system, $A(-4;-3)$, $B(1;-1)$ and $C(-1;3)$.

1. Determine D such that $ABCD$ is a parallelogram.

2. Determine its centre I in two ways.

3. With $E(2;1)$, determine K, intersection of (AB) and (CE).

4. Determine F such that $\overrightarrow{AF} = \frac{2}{3}\overrightarrow{AE}$. Are points D, F, B collinear?



Solution:

1. $D = A + C - B = (-6;1)$. 2. Midpoint of $[AC]$ and $[BD]$: $I(-5/2;0)$. 3. $(AB) : 2x - 5y - 7 = 0$; $(CE) : 2x + 3y - 7 = 0$. Subtracting gives $8y = 0$, then $x = 7/2$, so $K(7/2;0)$. 4. $\overrightarrow{AE} = (6;4)$, therefore $F = A + (4;8/3) = (0;-1/3)$. $\overrightarrow{DF} = (6;-4/3)$, $\overrightarrow{DB} = (7;-2)$; determinant $-12 + 28/3 = -8/3 \neq 0$. The three points are not collinear.

II Compare a vector method with a coordinate method

Example 12.2

In a non-degenerate triangle ABC, I is the midpoint of $[AB]$ and J the midpoint of $[AC]$.

1. Prove using vectors that $(IJ) \parallel (BC)$.

2. Check with $A(0;0)$, $B(6;0)$, $C(2;4)$.

3. For these coordinates, determine the equations of (IJ) and (BC), then the ratio IJ/BC.

Solution:

1. $\overrightarrow{IJ} = \overrightarrow{IA} + \overrightarrow{AJ} = \frac{1}{2}(\overrightarrow{BA} + \overrightarrow{AC}) = \frac{1}{2}\overrightarrow{BC}$. The vectors are non-zero and collinear.
2. $I(3;0)$, $J(1;2)$; $\overrightarrow{IJ} = (-2;2)$ and $\overrightarrow{BC} = (-4;4)$.
3. $(IJ) : x+y-3=0$; $(BC) : x+y-6=0$. Their direction vectors are collinear. $IJ = \sqrt{8}$, $BC = \sqrt{32}$, so $IJ/BC = 1/2$. The vector proof is general; the numerical check only shows this particular case.

Example 12.3

A figure seems to show two parallel lines. What is the difference between this observation and a proof?

Solution:

The figure suggests a conjecture and may be imprecise. A proof uses exact data and a property, for example a zero determinant of non-zero direction vectors. A conclusion must specify the objects concerned.

III Recognise a configuration and justify

Example 12.4: Constructed situation

In an orthonormal coordinate system, $A(0;0)$, $B(4;2)$, $C(3;4)$ and $D(-1;2)$.

1. Show that $ABCD$ is a parallelogram.

2. Calculate the four lengths and determine whether it is a rhombus.

3. Calculate AC^2, BD^2 and determine whether it is a rectangle.

4. Determine its centre and an equation of each diagonal.

Solution:

1. $\overrightarrow{AB} = (4;2) = \overrightarrow{DC}$: it is a parallelogram. 2. $AB = CD = \sqrt{20} = 2\sqrt{5}$, $BC = DA = \sqrt{5}$; the adjacent sides are not equal, so not a rhombus. 3. $AC^2 = 3^2+4^2 = 25$; $BD^2 = (-5)^2+0^2 = 25$. A parallelogram whose diagonals have the same length is a rectangle. We can also use $AB^2 + BC^2 = 20 + 5 = 25 = AC^2$ and the converse of Pythagoras' theorem. 4. The centre is $I(3/2;2)$. $(AC) : 4x - 3y = 0$; $(BD) : y = 2$. Their intersection gives I.

Guidance for the teacher

Comparing approaches: a proof by Pythagoras or by a property of the diagonals is enough. Do not impose the scalar product, which is not needed here. For the oral work, share the methods among groups, then ask for a common justification.